Activation Steering for Context-Balanced Synthetic Data

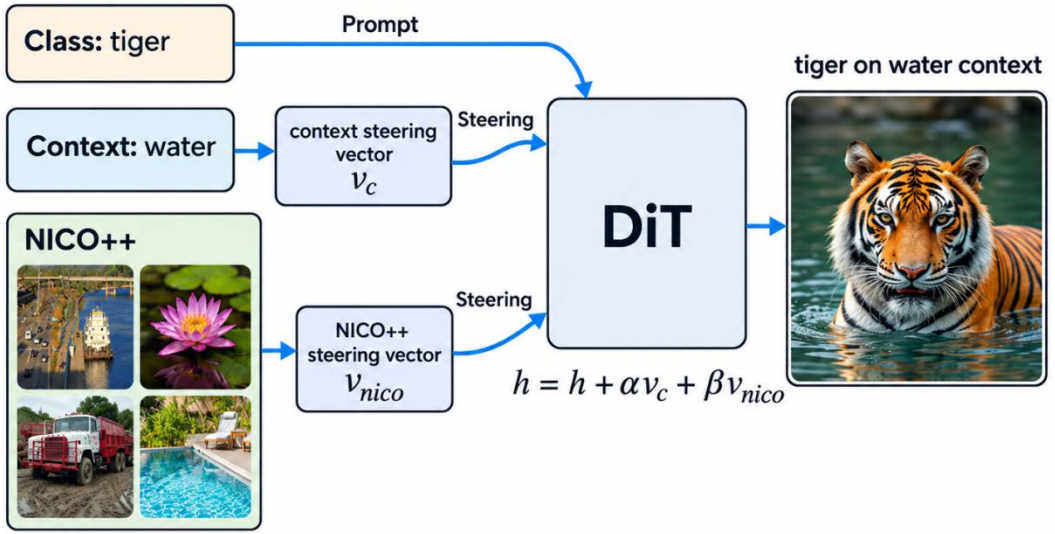
Can context-controlled synthetic images reduce the worst-group error of a classifier?

Dataset
NICO++: 60 classes, 6 contexts
Problem
Train groups are imbalanced: from **0** to **811** samples per class–context group
Baseline classifier
ResNet-50
Baseline result
Overall Accuracy = **86.52%**
Worst-Group Accuracy = **33.33%**

Idea: Generate rare groups using activation steering in DiT (SHIFT)

Hypothesis: after adding synthetic data to NICO++, the WGA of the classifier increases without increasing the size of the dataset, while maintaining overall accuracy.

1) Controlled generation



2) Add synthetic and train classifier

